

Module 03: Carrier Sense Multiple Access (CSMA) & Persistence

AKTU BCS603 Focus: Carrier sensing principles, 1-Persistent vs Non-Persistent vs p-Persistent strategies, and Propagation Delay collision window analysis.

CSMA Philosophy ("Listen Before Talk") & 3 Persistence Strategies

1. 1-Persistent CSMA

Greedy Transmission
Sense channel continuously

- If Idle: Transmit immediately ($P = 1$).
- If Busy: Keep sensing continuously.
The very instant line becomes idle, transmit!

High Collision Danger!
If 2+ nodes wait during busy period, both fire together when idle!

- Zero idle delay when line free
- Used in Classic IEEE 802.3 Ethernet

2. Non-Persistent CSMA

Polite Random Wait
Relinquishes immediate sensing

- If Idle: Transmit immediately.
- If Busy: Back off random time!
Do NOT continuously monitor line.

Drastically Reduces Collisions!
Different random waits prevent simultaneous firing

- Channel remains idle unnecessarily
- Lower channel efficiency at low load

3. p-Persistent CSMA

Probabilistic Slotted Compromise
Balances collision & idle wait

- If Idle: Transmit with probability p .
- With prob $(1 - p)$: Wait for next slot.
In next slot: Repeat test if still idle.

Mathematical Optimum
If N nodes ready, setting $p = 1/N$ avoids collisions

- Optimal trade-off under heavy load
- Used in IEEE 802.11 (CSMA/CA)

Why Collisions Still Occur in CSMA: The Propagation Delay (T_p) Window

Root Cause of CSMA Collisions: Signals take finite propagation time ($T_p = d/v$) to travel down the wire.

Collision Scenario: Host A senses channel (idle) and transmits at $t=0$. Before A's signal reaches distant Host B, B senses channel at $t = T_p - \epsilon$. B also sees channel as "idle" and transmits.

Vulnerable Window: CSMA Vulnerable Time = Exactly Propagation Delay (T_p). (ALOHA was $2T_{fr}$ or T_{fr} !)

1. Comparative Strategy Flow

- **1-Persistent:** "Aggressive". Channel idle $\rightarrow$ Transmit immediately. Channel busy $\rightarrow$ Keep listening, transmit at first instant of idle. *Pros:* Zero wasted channel time. *Cons:* High collision when busy period ends.
- **Non-Persistent:** "Polite". Channel idle $\rightarrow$ Transmit immediately. Channel busy $\rightarrow$ Wait random timer before re-sensing. *Pros:* Collision probability drops drastically. *Cons:* Channel sits idle while stations are in backoff sleep.
- **p-Persistent:** "Optimal Compromise". Slotted channel. If idle, transmit with probability p . With $(1-p)$, defer to next slot. Balances throughput and collision rate.

2. Throughput Comparison Curves (Kleinrock-Tobagi Model)

Channel load G ke badhne par throughput S :

Throughput S

-----> Offered Load G

AKTU Exam Question: "Discuss different carrier sense protocols. How are they different from collision protocols?"
me 1-Persistent, Non-Persistent aur p-Persistent ka flowchart aur T_p collision window explanation zaroor de!